

LatentWire: No-Text Candidate-Transfer Packets under Destructive Controls

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Abstract

Multi-model language systems usually communicate by emitting text or by sharing internal state such as KV caches. Text is portable but exposes source outputs; cache sharing is information-rich but moves and exposes dense model state. We study a deliberately narrower interface: can a frozen source model send a byte-scale packet that helps a frozen receiver on public multiple-choice reasoning tasks without sending source text, hidden states, KV cache, source logits, or labels? We present LatentWire, a packet protocol and evaluation framework with explicit byte accounting and destructive controls for source-choice, wrong-row, coordinate-shuffle, same-budget text, and cross-family shortcuts. On ARC-Challenge test, an 8-byte payload / 11-byte framed packet reaches 0.344 accuracy across 10 seeds, above target-only 0.265 and same-budget structured text 0.300. On OpenBookQA test, a 3-byte / 6-byte packet reaches 0.378 across 5 seeds, above target-only 0.276 but statistically indistinguishable from same-budget text under our paired-bootstrap gate. The decisive audit is also limiting: explicit source-index communication reaches 0.346 on ARC-Challenge and 0.378 on OpenBookQA, so the current positive is byte-scale candidate-hint transfer, not evidence synthesis beyond the source or a method that beats selected-candidate communication. We also report that a Phi-3 cross-family source fails and that simple cached connector repairs do not close the gap. The workshop contribution is a reproducible shape-restricted task-hint protocol, an unusually strict destructive-control suite, and byte/exposure accounting that separates measured packet objects from dense general-purpose state transfer. We do not claim solved latent communication, formal privacy, Slepian-Wolf/Wyner-Ziv rate-distortion guarantees, C2C superiority, or native GPU latency/HBM/energy wins.

1 Introduction

Future language-model systems will often use more than one model: a cheap model may screen candidates, a specialist may inspect hard cases, and a stronger receiver may combine evidence. The simplest interface is text. Recent cache-level systems instead communicate through model internals, such as fused or selectively shared KV caches (Fu et al., 2026; Shi et al., 2026; Ye et al., 2025). These approaches are powerful, but they couple communication to large hidden-state objects and serving substrates.

This paper asks whether a much smaller interface can carry useful source-side candidate evidence. In our setting, both models see the same public question and answer choices. The source performs private computation over those choices. The only communicated object is a short byte packet; the receiver never sees source text, source logits, source hidden vectors, KV cache, or the answer key. This is a side-information problem in the spirit of distributed source coding (Slepian & Wolf, 1973; Wyner & Ziv, 1976; Whang et al., 2021; Özyılkan et al., 2023): the receiver already has the public prompt and its own computation, so the packet should carry only the source evidence that is useful at the decoder. We use that framing as motivation, not as a formal rate-distortion theorem: this paper does not specify a full joint distribution, distortion measure, or converse. The scientific contribution is therefore not just

a packet row, but a controlled protocol for asking whether such rows survive the controls that usually create false positives.

Our strongest current result is deliberately scoped. On ARC-Challenge and OpenBookQA (Clark et al., 2018; Mihaylov et al., 2018), same-family Qwen2.5 packet transfer (Yang et al., 2025) improves over target-only. ARC-Challenge also clears the same-budget text gate, while OpenBookQA is statistically indistinguishable from same-budget text under the paired-bootstrap lower-bound gate. A strict Phi-3 source-family replacement (Abdin et al., 2024) fails, and additional connector repairs fail on the available TinyLlama/Qwen cached artifacts. This boundary matters: the paper is not claiming solved cross-family latent communication.

The contributions are:

- A byte-scale, content-private packet protocol and threat model for multiple-choice candidate transfer with no source text, hidden vector, KV cache, source-logit, or gold-label exposure.
- A strict evaluation suite, drawing on control-task and behavioral-testing methodology (Hewitt & Liang, 2019; Adebayo et al., 2018; McCoy et al., 2019; Ribeiro et al., 2020), that combines target-only, same-budget text, source-index/rank, coordinate-shuffle, wrong-row, same-source-choice, cross-family, and cached-connector controls.
- Controlled narrow positives on ARC-Challenge and OpenBookQA with seed repeats and paired uncertainty, accompanied by an explicit source-index audit showing that the current effect is mostly source-choice preserving.
- A failure-boundary analysis showing that Phi-3 cross-family transfer and simple cached connector repairs fail, so broad latent-language and cross-family claims are unsupported.
- A systems accounting artifact comparing raw, framed, cacheline-rounded packet sizes with dense KV/cache byte regimes while explicitly separating shape-restricted task hints from general-purpose state-transfer methods.

This is useful even with a narrow positive because it separates three questions that are often conflated: whether a compact private packet can help, whether it carries more than the source’s preferred answer, and whether it is cheaper or less exposing than dense state transfer. The current answer is positive for the first question, negative for the second, and accounting-only for the third. That scoped result is the paper’s intended workshop contribution.

Where the current result is useful. The current method is most plausible for same-family cascades, content-private specialist-to-router hints, and bandwidth-limited candidate reranking where sending source generations, logits, hidden states, or KV cache is undesirable. It is not yet a method for open-ended generation, heterogeneous cross-family collaboration, or reasoning synthesis beyond the source model’s selected candidate.

2 Content-private packet transfer

Let a public multiple-choice instance be $x = (q, c_1, \dots, c_m)$ with answer choices c_i . A source model S and receiver R both observe x , but only S computes the source-side private evidence $e_S(x)$. A packet encoder emits

$$z = E(e_S(x), x), \quad |z| \leq B \text{ bytes.}$$

The receiver predicts

$$\hat{y} = D_R(x, z),$$

where D_R may use public task features and the receiver’s own state, but may not access source text, source hidden states, source KV cache, source logits, or answer labels.

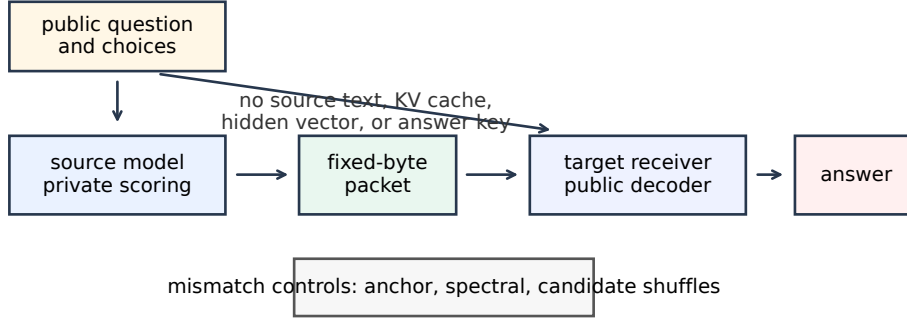


Figure 1: LatentWire protocol. Both sides receive the public task. The source privately scores candidates and emits only a fixed-byte packet. The receiver decodes the packet with a public basis and its own side information. Destructive controls scramble the shared coordinate system or candidate relation.

We evaluate a packet only if it beats three surfaces: target-only $D_R(x, \emptyset)$, a same-budget structured text control, and destructive controls that preserve superficial packet size but destroy the intended source-receiver relation. We also audit explicit source-index, source-label-text, source-rank-code, and entropy-matched random-index baselines. The same-budget structured text control is not the answer label “A/B/C/D” and not a direct source-index code: it sends the first B bytes of the source-selected public candidate string and decodes by lexical overlap with the public choices. Explicit source-index, source-label-text, source-rank, and source-score-style rows are reported separately as shortcut boundaries. We report seed repeats and paired bootstrap lower bounds. This matters because a packet can be useful while mostly preserving the source’s selected candidate; such a result is content-private candidate transfer, not proof of a shared latent language, reasoning beyond the source endpoint, or compression beyond a one-byte selected-candidate channel.

Threat model. The packet channel is content-private and source-state-private, but it is not identity-hiding, choice-hiding, or formally private in the differential-privacy, cryptographic, or information-theoretic secrecy sense. The sender may privately compute over the public prompt and answer choices, then transmit only a bounded byte string. The receiver may use that byte string, the public prompt, and its own computation. The receiver may not observe source generations, chain-of-thought, hidden states, KV cache, logits, score vectors, or labels. A passive packet observer sees only the packet bytes and framing, not source internals or text; however, the current packet leaks the source’s preferred candidate with 0.995 mean match on ARC-Challenge and 0.999 mean match on OpenBookQA. We therefore claim privacy of source content and dense state, not secrecy of source choice. A method fails the current workshop claim if its apparent gain is matched by a target-only cache effect, same-budget visible text, explicit source-index/rank/score communication, wrong-row transfer, or answer leakage.

Control suite. Table 1 lists the reviewer-facing controls used to interpret the reported rows. The suite is intentionally stronger than the headline claim: several controls are designed to collapse methods that only copy the source’s preferred answer choice.

3 Public-coordinate packets

The current ARC-Challenge implementation uses a public train-anchor chart. For each candidate, we compute a public relative-coordinate feature against a frozen set of training anchors, then project that feature into a low-frequency orthonormal DCT-II spectral basis. The packet is a sparse signed syndrome in this shared coordinate system. The receiver

Control family	What it tests	Current role
target-only / zero-source	target-cache or no-packet effects	required baseline
same-budget visible text	whether text at the same byte budget suffices	required baseline
source-index / rank / score	selected-candidate shortcuts	main claim boundary
wrong-row / row shuffle	row-specific source causality	destructive control
same-source-choice wrong-row	source-choice-only transfer	hardening control
anchor/bin/coordinate shuffle	shared-coordinate dependence	mechanism control
candidate	answer-order artifacts	destructive control
roll/derangement		
source-family substitution	cross-family generality	falsification control
cached connector repair	easy learned receiver rescue	negative diagnostic

Table 1: Control suite. The workshop claim requires packet utility to be interpreted through these controls; the current positive is explicitly bounded by source-index and source-family failures.

Step	Paper implementation
Source evidence	The source privately selects a candidate from the public choices using answer-key-forbidden source caches; no source text, logits, hidden states, KV cache, or labels cross the interface.
Public chart	Build candidate-pair text anchors from train rows; ARC-Challenge uses 384 public anchors and a 96-dimensional low-frequency DCT-II basis.
Source coordinates	Compute source candidate residuals in the public chart, project into the DCT basis, and keep the largest signed coordinates.
Packet	Encode each selected coordinate as index, sign, and quantized magnitude; ARC-Challenge uses 4 nonzero coordinates in 8 raw bytes / 11 framed bytes, and OpenBookQA uses 1 coordinate in 3 raw bytes / 6 framed bytes.
Receiver decode	Compute receiver-side candidate residuals in the same public basis, normalize candidate codes, and pick the candidate with highest cosine compatibility to the decoded packet vector.
Validation	Packet budgets and basis settings are selected before frozen test readout; source-index, label-text, rank-code, random-index, text, wrong-row, and shuffle controls are evaluated on the same frozen rows.

Table 2: Algorithm sketch for the reported packet rows. The table is intentionally explicit because the workshop claim depends on separating the no-text packet from direct source-index or source-label transmission.

123 scores candidate compatibility with the packet in the same public basis, combining the
124 packet evidence with receiver-side candidate information.

125 The design is related to relative representations (Moschella et al., 2023): anchors provide a
126 coordinate system less tied to raw hidden dimensions. Our use is narrower. We do not claim
127 zero-shot latent stitching or global representation alignment; we use a public basis to define
128 a tiny task packet. Random shared anchors work nearly as well as semantic anchors in
129 the current hashed ARC-Challenge implementation, so the correct interpretation is shared
130 coordinate agreement, not semantic anchor names.

131 OpenBookQA uses the same packet-evaluation contract at a smaller 3-byte budget. Across
132 both tasks, the source-choice cache is answer-key-forbidden: the source sees public questions
133 and choices, not labels. The packet is evaluated on frozen test slices after validation selection.

134 4 Main results

135 Table 3 and Figure 2 summarize the current positive evidence and the source-index audit.
136 The ARC-Challenge row uses 10 packet/control seeds and 2000 paired bootstrap samples
137 per seed on 1172 test examples. The OpenBookQA row uses 5 seeds on the 500-example
138 test slice. The source-index audit is generated by reconstructing the same packet rows and
139 adding explicit source-index, source-label-text, source-rank-code, and entropy-matched
140 random-index baselines.

141 **Reproducibility card.** All headline rows are frozen test-slice evaluations. The ARC-
142 Challenge row uses Qwen2.5-family source/target caches, 10 seeds, and 2000 bootstrap

Benchmark	split	bytes	seeds	src	packet	target	text	pkt-src	pkt-text
ARC-Challenge	test, 1172	8/11	10	0.346	0.344	0.265	0.300	-0.008	+0.025
OpenBookQA	test, 500	3/6	5	0.378	0.378	0.276	0.350	-0.006	+0.000

Table 3: Main audit rows. “Bytes” reports payload/framed bytes. “Src” is explicit answer-key-forbidden source-index accuracy before packet decoding. “Text” is the same-budget structured text control. “Pkt-src” and “pkt-text” are minimum paired-bootstrap 95% lower bounds for packet minus source-index and packet minus same-budget text. Packet-minus-target lower bounds are +0.038 on ARC-Challenge and +0.046 on OpenBookQA.

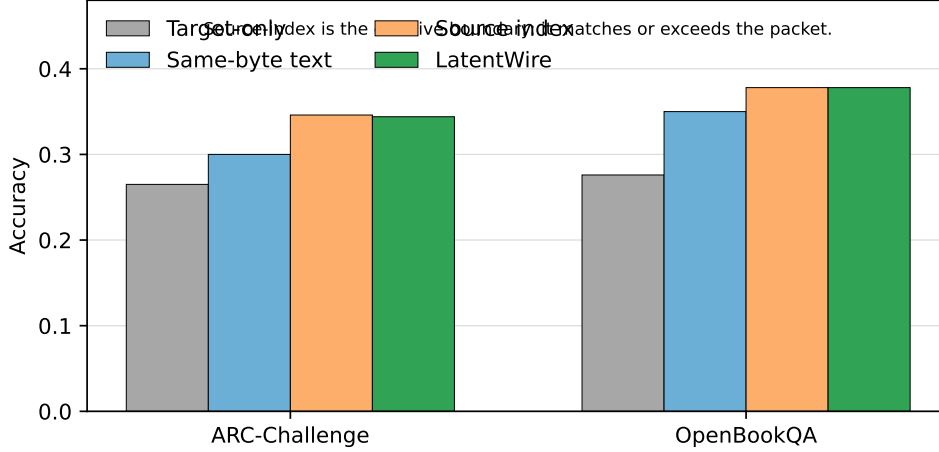


Figure 2: Packet accuracy against target-only, same-budget structured text, and explicit source-index controls. The source-index bar is the decisive boundary: it matches or exceeds the packet on both headline rows.

143 samples; the OpenBookQA row uses the same content-private contract with 5 seeds and 500
 144 bootstrap samples; the source-index audit reruns the same frozen prediction surfaces with
 145 explicit index/rank/label baselines. Exact input paths, hashes, scripts, and rerun status are
 146 listed in the artifact manifest and regenerated by the COLM.v3 review-packet script.

147 On ARC-Challenge, the matched Fourier/anchor-syndrome packet reaches 0.344 mean test
 148 accuracy with minimum seed accuracy 0.342. The answer-key-forbidden source index itself
 149 reaches 0.346, and packet predictions match the source-selected candidate at 0.995 mean
 150 rate across seeds. Anchor-ID shuffle, anchor-value shuffle, and spectral-bin permutation
 151 controls all fail in 0/10 seeds and collapse near target-only. The random shared-anchor
 152 diagnostic passes, which weakens any claim that named anchors are semantically special,
 153 but strengthens the claim that a shared public coordinate chart is the relevant mechanism.

154 On OpenBookQA, the 3-byte packet reaches 0.378–0.380 across seeds. The explicit source-
 155 index baseline reaches 0.378, and the packet follows the source index at 0.999 mean rate.
 156 The largest candidate-derangement control accuracy is 0.212. Although the packet is +0.028
 157 above same-budget structured text in mean, the paired-bootstrap lower bound is +0.000, so
 158 this is not a statistically supported text-baseline improvement. This second task reduces the
 159 risk that the result is only an ARC-Challenge artifact, but it confirms the same boundary:
 160 the current method is source-choice preserving and same-family, not cross-family latent
 161 reasoning.

162 The practical readout is an evidence ladder rather than a single win/loss row. The packets
 163 beat target-only on both headline rows; ARC-Challenge clears same-budget text, while
 164 OpenBookQA does not clear the paired lower-bound gate against text. Coordinate and
 165 candidate-destructive controls prevent treating the result as random packet regularization;
 166 explicit source-index/rank controls prevent overstating the mechanism. In framed-byte

Benchmark	seeds	pkt-target	pkt-text	pkt-source	provenance
ARC-Challenge	10	+0.038	+0.025	-0.008	frozen test, hash-audited
OpenBookQA	5	+0.046	+0.000	-0.006	frozen test, hash-audited

Table 4: Uncertainty summary for the main audit rows. Values are paired-bootstrap 95% lower bounds for packet accuracy minus the named comparator. The zero lower bound against same-budget text on OpenBookQA is intentionally visible: that row is a narrow secondary positive, not a strong text-baseline separation.

terms, the target-only lift is $0.079/11 = 0.0072$ accuracy per byte on ARC-Challenge and $0.102/6 = 0.0170$ accuracy per byte on OpenBookQA. Those rates are useful for comparing packet regimes, but they are not evidence that the current packet is better than a direct selected-candidate code.

Claim boundary. The current packet does not improve over explicit source-index communication: source-index reaches 0.346 on ARC-Challenge and 0.378 on OpenBookQA, while the packet reaches 0.344 and 0.378. A two-stage seed/item cluster bootstrap for packet minus source-index gives ARC-Challenge mean -0.0016 with 95% CI $[-0.0030, -0.0003]$ and OpenBookQA mean $+0.0004$ with 95% CI $[-0.0012, +0.0020]$. We therefore claim content-private fixed-byte candidate transfer under destructive controls, not compression beyond a selected-candidate channel.

Score-quantization diagnostic. The frozen headline ARC-Challenge/OpenBookQA source-choice caches store top choices, not raw calibrated score vectors, so source-score quantization is not a full headline baseline. We include a validation-only ARC-Challenge diagnostic with a 3-byte quantized source score distribution and receiver score side information. It reaches the same held-out accuracy as always using the source top label (0.389) and does not beat the top-label pair rule. This supports the same boundary: simple score sketches do not yet demonstrate information beyond source-choice transfer.

Breadth audits. We keep additional model and benchmark breadth in the reviewer packet rather than turning every diagnostic into a headline claim. A frozen benchmark-selection gate also records HellaSwag validation-1024 (Zellers et al., 2019) as a positive bounded validation diagnostic, CommonsenseQA (Talmor et al., 2019) as target-positive but same-byte-text saturated, and SciQ (Welbl et al., 2017) as a content-private bridge contract with no positive packet row yet. A separate latest-model emitter audit covers Qwen3.5, Gemma 4, Granite 3.3, and OLMo-2 source-packet behavior on a synthetic hidden-repair benchmark. These rows improve reviewer visibility into breadth and failure modes, but they are not used to strengthen the main ARC-Challenge/OpenBookQA claim.

5 Falsification and failure decomposition

The most important negative result is the Phi-3 source-family replacement. We keep the same ARC-Challenge packet budget, public basis, seed count, and bootstrap budget, but replace the Qwen source-choice cache with a Phi-3 source. The packet no longer passes. On the full ARC-Challenge test slice, the Phi-3 packet reaches 0.244 versus target-only 0.265. On the stricter Qwen-disagreement slice, where the alternate source and Qwen source disagree, the Phi-3 packet reaches 0.200 versus 0.340 for the Qwen-substituted packet.

The failure decomposition separates three causes: source endpoint quality, packet corruption, and common-feature mismatch. For Phi-3, packet predictions follow the Phi-3 source choice at 0.997, but the source itself is below the target. For TinyLlama, packet fidelity is also high, but the source-family mismatch causes the disagreement-slice packet to lose to Qwen-substituted packets. Across these rows, the packet mostly transports the source-selected candidate through the public coordinate basis. The source-index audit in Table 3 closes one obvious reviewer question: the current packet does not beat a direct selected-candidate

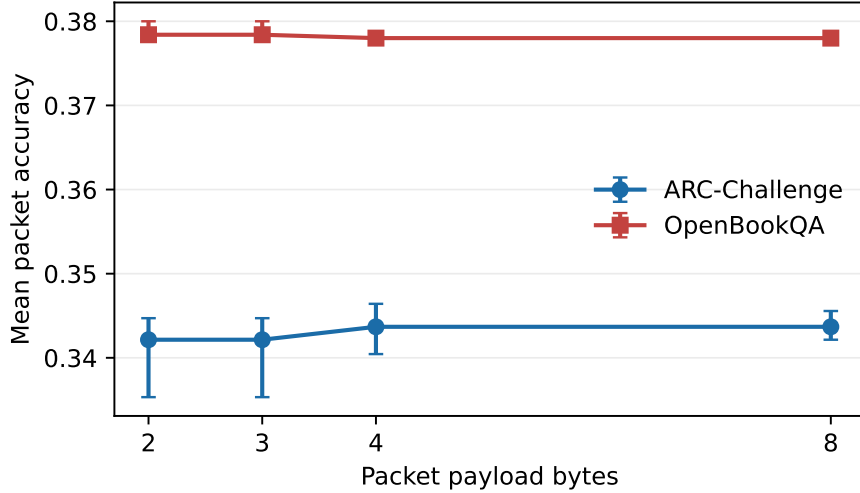


Figure 3: Payload-rate curve from the source-index audit. Packet payloads below 2 bytes are unsupported by the current sparse signed encoder; the 1-byte source-index baseline is reported separately in Table 3.

Source	surface	n	packet	target	text	ref. pkt	blocker
Phi-3	full test	1172	0.244	0.265	0.232	–	source quality
Phi-3	Qwen-disagree	833	0.200	0.273	0.203	0.340	source quality
TinyLlama	Qwen-disagree	473	0.269	0.268	0.258	0.317	family mismatch

Table 5: Cross-family diagnostics. “Ref. pkt” is the Qwen-substituted packet on disagreement rows. Validation-incomplete stronger-Qwen rows are kept out of the main paper and remain reviewer-packet diagnostics only.

code. The next research gate is therefore a richer common-feature connector or calibrated source-score packet that can improve beyond packet-only source choice.

We also tested cached learned repairs on the available local artifacts. A candidate-syndrome connector that sees packet and score-shape features reaches 0.288 versus 0.317 for Qwen-substituted packets on frozen TinyLlama/Qwen disagreement rows. A one-hidden-layer MLP over cached TinyLlama hidden/query summaries reaches 0.232, below Qwen-substituted packets, cached Tiny packets, target-only, and same-budget text. These negatives are useful: simple cached score geometry and mean hidden/query summaries do not explain the remaining oracle headroom.

6 Systems boundary

The systems contribution is byte/exposure accounting, not native serving throughput. Figure 4 compares framed packet bytes with one-token KV/cache-state accounting floors from neighboring systems and quantization methods. LatentWire sends 6–11 framed bytes on the two headline rows. A one-token 1-bit-per-KV-element accounting floor is 768 bytes in our model-configuration accounting; the largest headline packet is therefore 69.8x smaller as a message object than that floor. This is not a function-equivalence claim. C2C-style cache transfer, KVComm/KVCOMM-style selective sharing, and serving substrates such as vLLM/PagedAttention and SGLang operate on KV/cache objects rather than shape-restricted task hints (Fu et al., 2026; Shi et al., 2026; Ye et al., 2025; Kwon et al., 2023; Zheng et al., 2024). KV quantization work reduces cache bytes but still communicates or stores model state (Zandieh et al., 2024; Liu et al., 2024; Hooper et al., 2024; Zandieh et al., 2026).

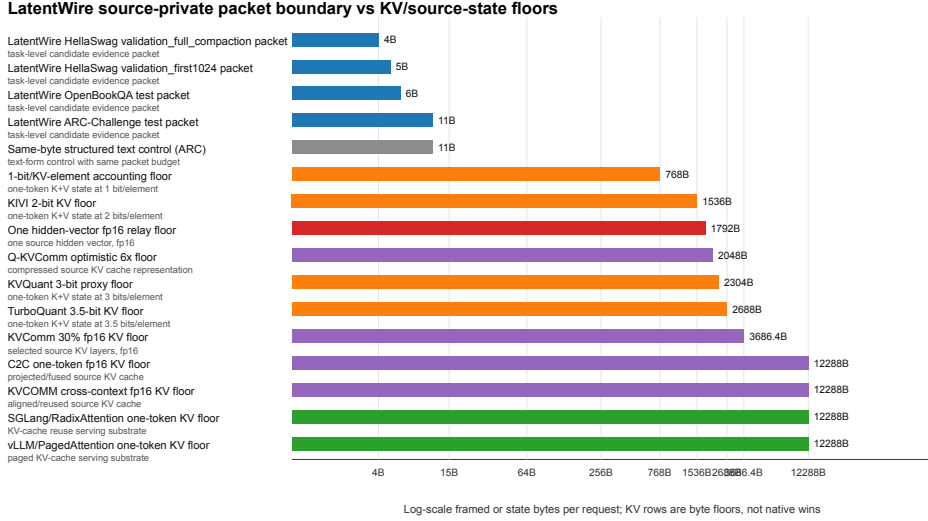


Figure 4: Boundary-byte accounting. Packet rows are measured task-packet objects. KV/cache rows are lower-bound floors or neighboring serving substrates, not native speed comparisons.

Method	raw	framed	text	KV/hidden	claim status
LatentWire OpenBookQA packet	3B	6B	no	no	local positive
LatentWire ARC-Challenge packet	8B	11B	no	no	local positive
Same-byte text control	8B	11B	yes	no	control only
1-bit/KV-element floor	768B	768B	no	KV	floor only
KIVI 2-bit KV floor	1536B	1536B	no	KV	floor only
C2C one-token KV floor	12288B	12288B	no	KV	native pending

Table 6: Exposure classes. Packet rows are measured/accounted local communication objects. KV/cache rows are analytical byte floors or native-pending baselines. The paper-ready systems claim is object size and source-exposure accounting, not GPU throughput, TTFT, TPOT, HBM, energy, or C2C superiority.

229 We label packet-object bytes as measured/accounted local artifacts and dense-state rows as
 230 analytical or literature-derived byte floors. This distinction is important: the current data
 231 support privacy/exposure and byte-budget comparisons, but not claims about GPU latency,
 232 HBM traffic, energy, or serving throughput.

233 7 Related work

234 **Latent and relative representation communication.** Relative representations use similar-
 235 ties to anchors to enable latent-space communication under invariances (Moschella et al.,
 236 2023). Sparse autoencoder work motivates interpretable feature coordinates (Huben et al.,
 237 2024), while feature-space universality work suggests that some features can be related
 238 across models (Lan et al., 2024). Crosscoders, transcoders, and activation-transfer methods
 239 are natural future sources of interpretable packet atoms or dense communication baselines.
 240 SAEbench provides evaluation machinery for sparse autoencoders (Karvonen et al., 2025),
 241 but these works do not by themselves provide a byte-counted packet protocol or task-level
 242 evidence transfer result under source-choice controls.

243 **Embedding communication and steganography.** CIPHER-style embedding communica-
 244 tion and embedding-space steganography are closer in spirit than text prompting because
 245 they communicate through non-natural-language vectors or geometry (Pham et al., 2024;
 246 Westphal et al., 2026). LatentWire differs in bandwidth and interface: the communicated

Regime	Example	Object crossing interface	Difference from LatentWire
dense cache fusion	C2C	projected source KV cache	high-bandwidth, source-state exposing
selective KV sharing KV compression	KV-cache comm. KV quantizers	selected KV pairs/layers compressed KV/cache state	cache-level, native pending shrinks dense state, not content-private task packets
serving substrate	vLLM/SGLang	KV reuse and memory management	runtime baseline, not communication protocol
prompt/prefix compression	prefix/gist tokens	prompt-like tokens or parameters	not content-private model-to-model evidence
local shortcut base-lines	source-index/rank/score	selected answer or score sketch	must be beaten for stronger claims

Table 7: Related-work and baseline matrix. LatentWire is positioned as low-byte content-private packet transfer with destructive controls, not as a replacement for dense cache fusion or serving-runtime optimization.

object is a discrete packet decoded through a public chart, not a continuous embedding stream inserted into a model. The steganography literature is also a reminder that “private” terminology should be quantitative; here we make only a content/state exposure claim and explicitly report source-choice leakage.

Destructive controls. Our control suite is comprehensive, but it is not novel as a methodology in kind. It borrows from control tasks, saliency sanity checks, adversarial/heuristic dataset controls, and behavioral tests (Hewitt & Liang, 2019; Adebayo et al., 2018; McCoy et al., 2019; Ribeiro et al., 2020). The paper’s claim is the application and aggregation of those controls for byte-scale inter-model packets, not invention of falsification as a new evaluation paradigm.

Prompt compression and learned prefixes. Prefix tuning and gist tokens learn continuous or discrete prompt-like interfaces within language models (Li & Liang, 2021; Mu et al., 2023); text-compression systems such as LLMingua and LongLLMingua are stronger visible-text baselines for future work (Jiang et al., 2023a;b). LatentWire differs because the communicated object is an opaque byte packet decoded by a public receiver, not a learned soft prompt injected into one target or a compressed visible prompt. Query-bottleneck connectors such as BLIP-2’s Q-Former show that learned bottlenecks can bridge frozen model families across modalities (Li et al., 2023); our failed cached hidden/query summaries suggest that a true tokenwise query-bottleneck is the right future branch, but it is not closed here.

Cache communication and systems. Cache-to-cache, KVComm, and KVCOMM communicate richer internal state and are the closest competitors for model-to-model communication (Fu et al., 2026; Shi et al., 2026; Ye et al., 2025). vLLM and SGLang improve serving through KV-cache management and reuse (Kwon et al., 2023; Zheng et al., 2024). KV quantizers such as QJL, KIVI, KVQuant, and TurboQuant reduce state cost but remain state-transfer or state-storage techniques (Zandieh et al., 2024; Liu et al., 2024; Hooper et al., 2024; Zandieh et al., 2026). Table 7 summarizes how these baselines differ from LatentWire. Our current systems result is a boundary artifact against these state objects, not a native throughput win.

8 Limitations and next gate

The main limitation is scope. The positive results are same-family/source-cache results on public multiple-choice tasks, and the packet largely preserves the source’s selected candidate. We therefore do not claim to beat explicit source-index communication, source-score quantization, or calibrated source-choice baselines; those are required before a stronger compression claim. OpenBookQA uses only 5 seeds, so small separations from same-budget text should be treated as unresolved. More importantly, 4-way multiple-choice tasks have a low answer-channel ceiling: a 1-byte source-index code is already a strong competitor. A higher-entropy task, calibration/abstention metric, or multi-step downstream use is needed to demonstrate packet information beyond selected-candidate transport. Phi-3 cross-family replacement fails, TinyLlama common-feature repair fails, the validation-only

score-sketch diagnostic does not beat source-label transfer, and a true cross-family learned query-bottleneck remains future work. We also do not claim formal Slepian–Wolf/Wyner–Ziv guarantees, formal privacy, or superiority over C2C, KVComm/KVCOMM, QJL, KIVI, TurboQuant, vLLM, or SGLang in native serving metrics. Those rows require NVIDIA hardware and matched implementations.

The next exact gate for an ICLR-strength claim is a frozen-endpoint tokenwise connector with an explicit byte or query budget, tested against C2C/KV-style and structured-text baselines, with source-destroying controls and strict same-family versus cross-family separation. The current strongest claim is narrower: fixed-byte content-private packets can transfer useful same-family candidate evidence on ARC-Challenge and OpenBookQA, and the accompanying control ladder clarifies where the method stops working.

9 Conclusion

LatentWire shows that a tiny content-private candidate packet can improve frozen receiver decisions on two public QA benchmarks under strict same-family controls. The same evidence also draws a hard boundary: the current method is mostly source-choice preserving, is not a universal latent language, and does not yet solve cross-family communication. This combination of positive packet transfer, negative controls, and systems byte/exposure accounting is the paper’s core.

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A Reviewer-facing claim audit

Table 8 states the claim boundary used to write the main paper. The full generated packet is in [results/latentwire.colm.v3.review.packet.20260505/](#).

Claim	Status	Evidence	Required wording
content-private packet protocol	supported	packet rows and controls	protocol/evaluation contribution
narrow same-family packet utility	supported but narrow	ARC-Challenge and Open-BookQA rows	source-candidate transfer
shortcut collapses are common	supported	source-index and wrong-row audits	claim boundary, not failure-only story
beats C2C or dense KV transfer	unsupported	no native matched C2C row	do not claim
native GPU/HBM/energy win	unsupported	no native NVIDIA measurement	future runbook only
broad latent/cross-family communication	unsupported	Phi-3 and connector failures	ICLR future goal

Table 8: Claim audit used for COLM.v3 integration. Each main-text claim must map to one supported or supported-but-narrow row.

B Artifact manifest

- ARC-Challenge headline: [colm_final/evidence/results/source_private_arc_challenge_fourier_anchor_syndrome_gate_20260502.budget8_10seed.b2000](#).
- Phi-3 falsification: [colm_final/evidence/results/source_private_arc_challenge_source_family_cache_falsification.20260502_phi3.cpu.budget8_10seed.b2000](#).
- Cross-family decomposition: [colm_final/evidence/results/source_private_arc_cross_family_failure_decomposition.20260502](#).
- Systems boundary: [colm_final/evidence/results/source_private_systems_boundary_figure_table.20260502](#).
- OpenBookQA headline: [colm_final/evidence/results/source_private_openbookqa_seed_stability_20260501_qwen05_hashed_test.3b](#).
- Source-index audit: [colm_final/evidence/results/source_private_colm_acceptance_baselines.20260502](#).
- Aggregate source-index CI: [results/source_private_colm_acceptance_baselines.20260502/aggregate_source_index.ci.md](#).
- Score-sketch diagnostic: [results/source_private_arc_score_fusion_packet_probe.20260503.qwen05.qwen3.validation](#).
- COLM.v3 review packet: [results/latentwire.colm.v3.review.packet.20260505](#).

C Reproducibility summary

The artifact bundle records exact shell commands, input hashes, output hashes, and rerun results. The paper rows are produced from cached, answer-key-forbidden source-choice artifacts with the following settings:

Row	script	seeds	key settings
ARC-Challenge	Fourier/anchor syndrome gate	10	8B, 2000 bootstraps
OpenBookQA	seed-stability gate	5	3B, hashed features, 500 bootstraps
Source-index audit	acceptance baseline audit	10/5	1000 bootstraps, 2/3/4/8B rate curve
Aggregate pkt-src CI	aggregate CI script	10/5	10000 two-stage seed/item bootstraps
Phi-3	source-family cache falsification	10	8B, cached Phi-3 source
Systems	systems boundary table	–	byte/exposure accounting only

Frozen artifacts are hash-verified. In the final local pass, OpenBookQA, systems, and failure-decomposition reruns completed; the more expensive full ARC-Challenge/Phi-3 reruns are represented by frozen hashes unless rerun offline. Rerun JSON files are metric-oriented rather than byte-identical because they contain creation times and some artifacts include measured local latencies.

D Per-seed audit rows

Table 9 exposes the per-seed test rows behind the main source-index audit. The important reviewer readout is sign consistency: packet-minus-source-index is never a positive lower-bound gate on either benchmark, while ARC-Challenge consistently clears same-budget text and OpenBookQA does not.

Benchmark	seed	packet	source-index	text	follows source	CI pkt-source
ARC-Challenge	47	0.342	0.346	0.300	0.997	-0.007
ARC-Challenge	53	0.346	0.346	0.300	0.995	-0.004
ARC-Challenge	59	0.343	0.346	0.300	0.990	-0.008
ARC-Challenge	61	0.345	0.346	0.300	0.997	-0.003
ARC-Challenge	67	0.343	0.346	0.300	0.995	-0.007
ARC-Challenge	71	0.344	0.346	0.300	0.995	-0.005
ARC-Challenge	73	0.344	0.346	0.300	0.997	-0.005
ARC-Challenge	79	0.343	0.346	0.300	0.997	-0.006
ARC-Challenge	83	0.346	0.346	0.300	0.996	-0.003
ARC-Challenge	89	0.345	0.346	0.300	0.995	-0.005
OpenBookQA	47	0.378	0.378	0.350	1.000	0.000
OpenBookQA	53	0.378	0.378	0.350	0.996	-0.006
OpenBookQA	59	0.378	0.378	0.350	1.000	0.000
OpenBookQA	61	0.380	0.378	0.350	0.998	0.000
OpenBookQA	67	0.378	0.378	0.350	1.000	0.000

Table 9: Per-seed test audit. “CI pkt-source” is the per-seed paired-bootstrap 95% lower bound for packet minus explicit source-index. Full CSV is stored in [results/source_private.colm.acceptance.baselines.20260502/per_seed_baseline_metrics.csv](#).

Workshop checklist. The current paper is ready for reviewer circulation only after the generated COLM_v3 packet and the PDF agree on: (1) the main claim, (2) the content-private threat model, (3) all main table values, (4) measured-versus-estimated systems labels, and (5) unsupported claims. The repository command

```
./venv_arm64/bin/python scripts/build_latentwire_colm_v3_review_packet.py
```

regenerates the claim audit, table/figure inventory, systems ledger, artifact manifest, and NVIDIA native runbook from tracked source artifacts.

E Lay explanation of the main experiment

The experiment asks whether one model can send a tiny hint to another model. Both models see the same question and answer choices. The sender chooses a compressed coordinate hint, only a few bytes long. The receiver uses that hint to choose an answer. If both sides agree on the coordinate system, the hint helps. If we scramble the coordinate system, the hint stops helping. That is evidence for a real shared packet interface on the tested same-family setting, while the failed Phi-3 run shows the interface is not yet general across model families.